# Bing Gao

Computational Mechanics Student | CFD & Aerodynamics

Paris, France | binggao1230@gmail.com | binggao1230 | binggao1230 | binggao.dev

## ENGINEERING EXPERIENCE

**Vinci Eco Drive**, *Aerodynamics Engineer* — Sept 2025 – present
*Project: Formula Student race car — cooling & aerodynamic design* — *Paris, France*

- **Cooling Design & Calculations:** Designed motor, inverter and battery cooling systems; selected radiators, fans, pumps and piping using heat-transfer, energy-balance and pressure-loss calculations for flow requirements and cooling capacity.
- **CAD & Aerodynamics:** Modeled sidepods, cooling ducts, deflectors and the rear wing in 3DEXPERIENCE; assessed airflow paths and component placement to balance heat rejection and aerodynamic drag.
- **OpenFOAM Simulation:** Simulated cooling airflow and external aerodynamics; compared flow rate, pressure drop, temperature and drag across design variants, using the results to refine cooling-duct and aerodynamic geometry.

**Felisiak Ingénierie & Développement**, *3D Modeling Intern* — Apr 2026 – Aug 2026
*Project: European Space Agency Space Rider — spacecraft reconstruction* — *Paris, France*

- **Spacecraft Reconstruction:** Reconstructed Space Rider's main body and key components from public engineering drawings, technical manuals and reference images; completed the assembled 3D model in Blender.
- **Python Geometry Assessment:** Automated batch sampling of mesh vertices, contours and cross-sections with Blender API scripts; calculated maximum deviation and root-mean-square error against reference drawings.

## SOFTWARE ENGINEERING EXPERIENCE

**Inkeverse**, *Mobile Application Development Manager* — July 2016 – Aug 2025
*Project: LaoYou — TikTok-like live-streaming social platform* — *Beijing, China*

- **Product & Technical Leadership:** Wrote over 70% of the initial core code for a platform that grew to 2M+ users; led technical design, task planning, code reviews, releases and long-term maintenance.
- **Full-Stack, Multi-Platform & AI-Assisted Development:** Developed Android and iOS applications, integrated React Native, built mini-programs with Taro, and developed back-end services with Ruby on Rails; used Cursor, Claude Code, and Figma integrations through the MCP for requirement analysis, interface implementation, refactoring, and testing.
- **Real-Time Systems & Release Engineering:** Developed live-room architecture, audio/video processing, real-time gift rendering, and voiceprint-recognition features; built Ruby-based automation for testing, builds, staged rollouts, and app-store delivery, and implemented hotfix mechanisms, code obfuscation, SSL Pinning, and production-incident troubleshooting.

**Google Developer Group**, *Core Organizer* — Nov 2015 – Aug 2025 — *Beijing, China*

- **Community Leadership:** Delivered 5+ technical events annually with 1k attendees each; coordinated topics, speakers and event delivery with engineers from Google and other tech companies.
- **Open-Source Contributions:** Organized sessions on Android, Kotlin, Flutter, TensorFlow and Google Cloud; contributed to open-source development, technical-content localization and developer support.

**Creating EV Technology**, *Android Engineer Intern* — Aug 2015 – July 2016
*Project: ClassBox — university course management platform* — *Beijing, China*

- **Android & Back-End:** Contributed to 11 ClassBox 7.0 releases, implementing course management, Coursera integration, calendars, widgets, in-app updates, WebView, and contributed to APIs for course mapping, account binding and e-commerce.

## EDUCATION

**ESILV**, *Diplôme d'ingénieur (Master's level), Modelling & Computational Mechanics* — Sept 2025 – June 2027 — *Paris, France*

- **Coursework Focus:** CFD, FEM, Fluid Flow Analysis, Aeroelasticity, Structural Optimization

**Changchun University of Science and Technology (CUST)**, *Bachelor of Science, Computer Science* — Sept 2012 – July 2016 — *Changchun, China*

- **Coursework Focus:** Numerical Analysis, Mathematical Modeling, Data Structures & Algorithms
- **Honors:** National Scholarship (top 0.2%); 1st Class Scholarship ×5; 1st Prizes in Mathematical Modeling and ACM competitions.

## SKILLS & LANGUAGES

**Mechanical Engineering:** OpenFOAM, Ansys Fluent, STAR-CCM+, Abaqus, CATIA/3DEXPERIENCE, SolidWorks, Blender

**Programming:** Python, C/C++, MATLAB/Simulink, Java/Kotlin, Ruby, JavaScript/TypeScript, SQL, Git, Linux

**Languages:** Chinese (native), English (C1), French (B2)

**Interests:** Black-run snowboarding, amateur racing, animal rescue volunteering, short-film filmmaking